



Annex OMP 14-B

Qualification main steps and contractual principles

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1. Purpose of Annex B

Although the Qualification process needs to be tailored to fulfil the specific needs of each OCCAR-managed Programme, it shall nevertheless adhere to key principles during the various steps of the process. The steps described hereafter in this Annex B may include additional sub-steps, which may partially overlap or be complemented by additional sub-steps as required to best suit the individual programme requirements.

2. Qualification-related activities prior to Contract award

It is essential that PSs achieve a common and unambiguous set of the Programme Requirements.

Prior to contract award, OCCAR-EA, in coordination with the PSs, must define the requirements for the Product (to be procured), these must be in line with the High-Level Objectives, legal, qualification, certification, acceptance and other criteria. Those requirements may have already been established before integrating the programme into OCCAR-EA.

Proper Requirements Management is essential for a successful management of Qualification.

OCCAR-EA should, in coordination with the PSs, harmonise with the potential future Contractor preliminary Qualification and Certification plans outlining by which means the Contractor will demonstrate that the Product Design meets the Product Specification as well as the structure of the body of evidence necessary to qualify and/or certify. The incorporation of such plans when the Contract enters into force is a best practice to reduce the risks of future requests for trials stemming from the PSs and not covered within the scope of the Contract. For example, an additional risk-reduction best practice is the assessment of competences and maturity of the Design Organization in order to adjust the adequate level of involvement of Nations during the development phase (mandatory for the success of Air programmes).

For COTS/MOTS/NDI projects, the approval of requirements (as described below in §3 Step 1) is performed during the tendering stage, before granting a Contract.

3. Qualification-related activities within the Contract

Contractual requirements shall include specific provisions to cover Qualification and Certification activities.

The Qualification and Certification requirements should consider synergies stemming from a mixed civilian and military (dual use) Qualification and Certification process.

The Contractor shall be responsible for successfully achieving the Qualified Design within Qualification of the Product or system as specified in the Contract.

The Qualification process for Governments Furnished Equipment (GFE) will be limited to the validation of their interface with the system unless otherwise agreed upon in writing between the Contracting Authority (CA) and the Contractor.

Qualification-related activities within the Contract follow below steps:

- Step 1: Definition and Approval of the Product Specification and Product Design

Product Specification Approval Sub-Step: The Contracting Authority (CA) defines the requirements (which may be a specification) to be met by the Product (including its support systems as well as its interfaces to other connected systems and systems of higher level in the case of a system of systems). The Contractor develops the Product Specification tailored to meet the requirements for the Product. Once approved by the CA¹, the Product Specification becomes the basis for the Contractor to finalise the Product Design. The Product Specification remains the Contractor's document and does not relieve the Contractor of meeting the contractual requirements.

Product Design Approval Sub-Step: Using the Product Specification a Product Design solution shall be developed by the Contractor and presented to the CA for evaluation and reviews. Design reviews may be required during this sub-step (e.g. System Requirements Review (SRR), System Design Reviews, Test Readiness Review (TRR), Preliminary Design Review (PDR), Critical Design Review (CDR)). If a Product Design contains sub-contractor items/part, it is essential that the review be conducted against Contract requirements and not against an industrial specification of the sub-contractor or Contractor.

The performance of a Test Readiness Review (TRR) is not a mandatory step in the process but is highly encouraged.

It remains the responsibility of the Contractor to ensure that the Product Design meets the contractual requirements.

When the Product Design is considered to be compliant with the Product Specification and it meets the contractual requirements the Product Design will be approved by the CA, then Step 1 is complete. The deliverable of this Sub-Step is the Product Design documentation.

- Step 2: Planning for Qualification

The aim of the Qualification process is to demonstrate the Product Design's ability to fulfil the specified contractual requirements (Product Specification).

The Qualification Programme Plan (QPP) to be developed by the Contractor explains how the Product Design will be qualified against the contractual requirements. The QPP follows an iterative harmonization process.

The QPP should also consider any applicable test procedures required by the Certification authorities and Individual Acceptance seeking synergies and thus avoiding duplication of tests. The Contractor should present the QPP to the CA for review and approval (this may require several iterations).

Although a large proportion of the Qualification will be managed in the frame of the Contract to the Contractor, Qualification should also consider other processes which are performed under the responsibility of each PS and which may be based on further testing in operational conditions (e.g. approval of entry into service, operational trials).

¹ Specification approval may be omitted if the OCCAR-managed Programme contains no development phase.

In cases where a CQO (or equivalent formal structure has been established), the CQO will review the QPP to verify in detail the proposed compliance demonstrations and, when relevant, define derived requirements, boundary conditions and pass/fail criteria for tests and evaluations. The CQO should provide formal recommendations to the Programme Committee² (PC).

It is essential for a realistic and achievable QPP to ensure that all stakeholders are aware of their required commitment during Qualification (GFX and human resources). Special attention is also required to address participation of some organisations (e.g. Certification Authorities). With respect to the commitment from the PSs regarding the provision of required national resources, the QPP needs to be endorsed by the PC prior to approval by the CA. The ProgD should include as much detail on the required commitment from the PSs' or the mechanism to secure this commitment.

The deliverable of Step 2 is the approved QPP.

▪ Step 3: Conducting Qualification

Qualification is performed and monitored against the QPP.

The approval process of the Qualification Test Plan (QTP), Requests For Deviations (RFDs) and Qualification Reports (QRs) shall be included as a contract requirement (for the Qualification Programme under the contract).

The QTRs, RFDs and final QRs will be evaluated and approved by the PM (assisted by national experts as appropriate) or by the CQO if established (in which case the CQO shall provide a formal recommendation to the PM). The Qualification is considered to be achieved when all four following conditions are fulfilled:

- Whenever a certification is a pre-requisite to grant qualification (including among others aeronautical products, ships, land-based vehicles), a certification is available;
- All Compliance Demonstrations have been performed according to the QPP/QTP specifications, and the results/evidence have been documented in the QTRs/QR;
- The results/evidences demonstrate achievement of pass criteria; or the corresponding RFDs have been approved by the PM;
- The final QR is approved in accordance with the contract.

A Qualified Design will result when Qualification has been achieved. The deliverable of this step 3 is the Qualified Design documentation.

Otherwise, in the event these conditions have not been met in accordance with the contract the Contractor may be required to perform new or additional tests or design modifications (i.e. it may be necessary to go back to Step 1 – definition and approval of Product Specification) in order to verify and validate the Product Design and achieve Qualification.

As a general principle, the availability of a Qualified Design is a pre-requisite to conduct Individual Acceptances, unless otherwise settled with the PSs.

²The Programme Committee may delegate authority for approval to the OCCAR Programme Manager.